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ASSIGNMENT 4

FINAL PROJECT

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Table of Contents

PROJECT OVERVIEW	3
SETUP & RUNNING	3
SYSTEM UI OVERVIEW	4
SYSTEM ALGORITHMIC LOGIC (A1-A3)	6
CONCLUSION	8
PEER REVIEW	8

PROJECT OVERVIEW

This Proof of Concept (POC) presents an **AI Task Automation System** designed to simulate intelligent decision-making in an industrial environment. The system autonomously assigns tasks, manages workflows, monitors machines, optimizes resources, and detects anomalies.

The solution integrates:

- **A1 – Knowledge Representation:** Tasks, workers, machines, resources, workflows, and alerts are explicitly modeled as structured states and actions.
- **A2 – State Space Search:** The AI agent transitions between system states by executing valid actions that improve overall system performance.
- **A3 – PEAS Model:** The agent's behavior is formally defined through performance measures, environment, actuators, and sensors.

The POC demonstrates how an AI agent can **observe, decide, and act** in a dynamic system through a step-by-step simulation.

SETUP & RUNNING

The POC is implemented using **Python** and runs as an interactive simulation.

Setup steps:

1. Ensure Python 3.9+ is installed.
2. Install dependencies:

```
pip install -r requirements.txt
```

3. Run the system:

```
python main.py
```

If this "WARNING: The script flask.exe is not in a PATH" occurred.

Run this:

```
python -m flask --app main run
```

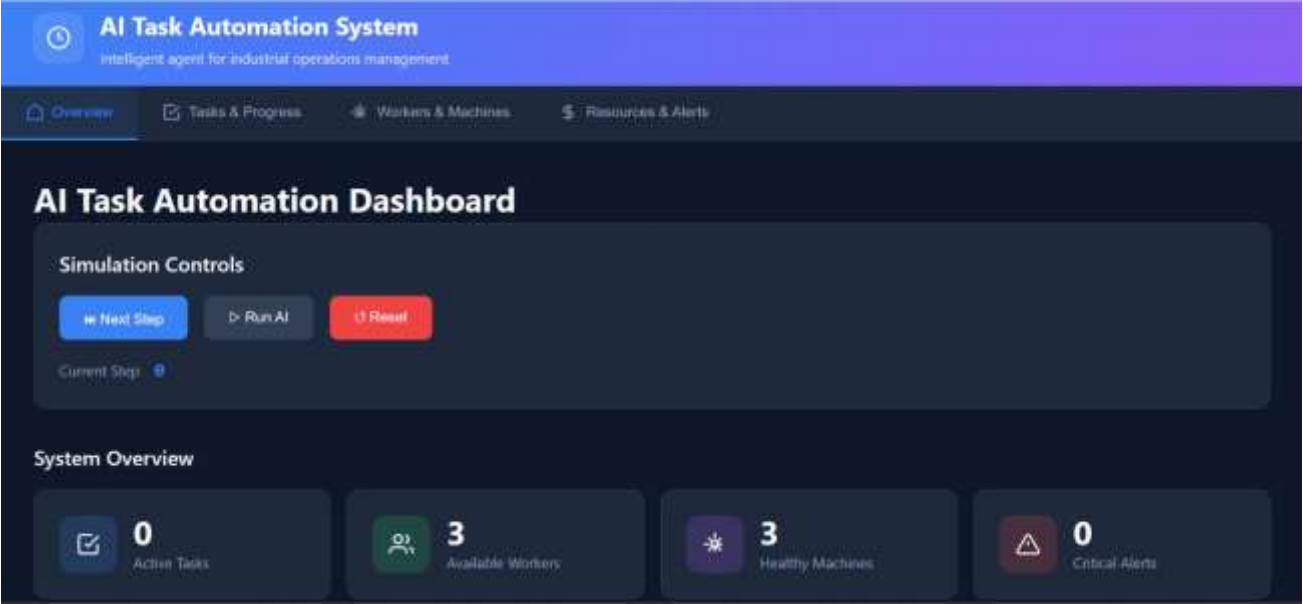
4. Access the dashboard at: ****http://localhost:5000****

Upon execution, the system initializes tasks that are retrieved from dataset, workers, and machines, and allows the user to run the simulation step-by-step or automatically, enabling clear observation of AI decision-making.

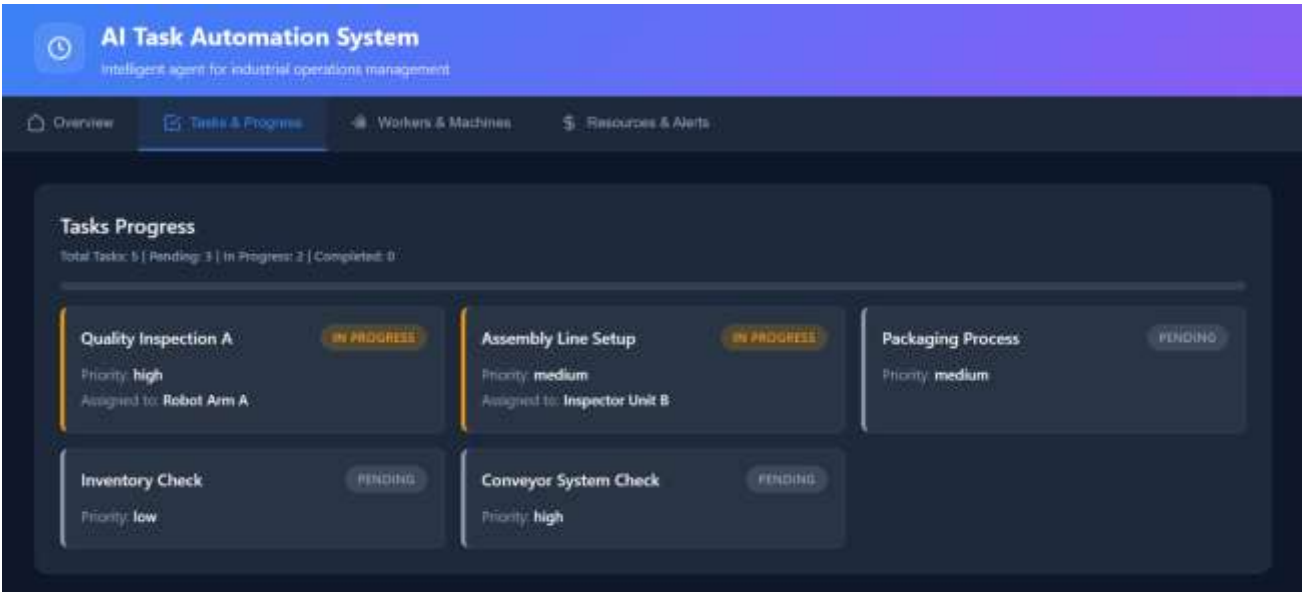
SYSTEM UI OVERVIEW

The system user interface visualizes the AI agent’s behavior in real time:

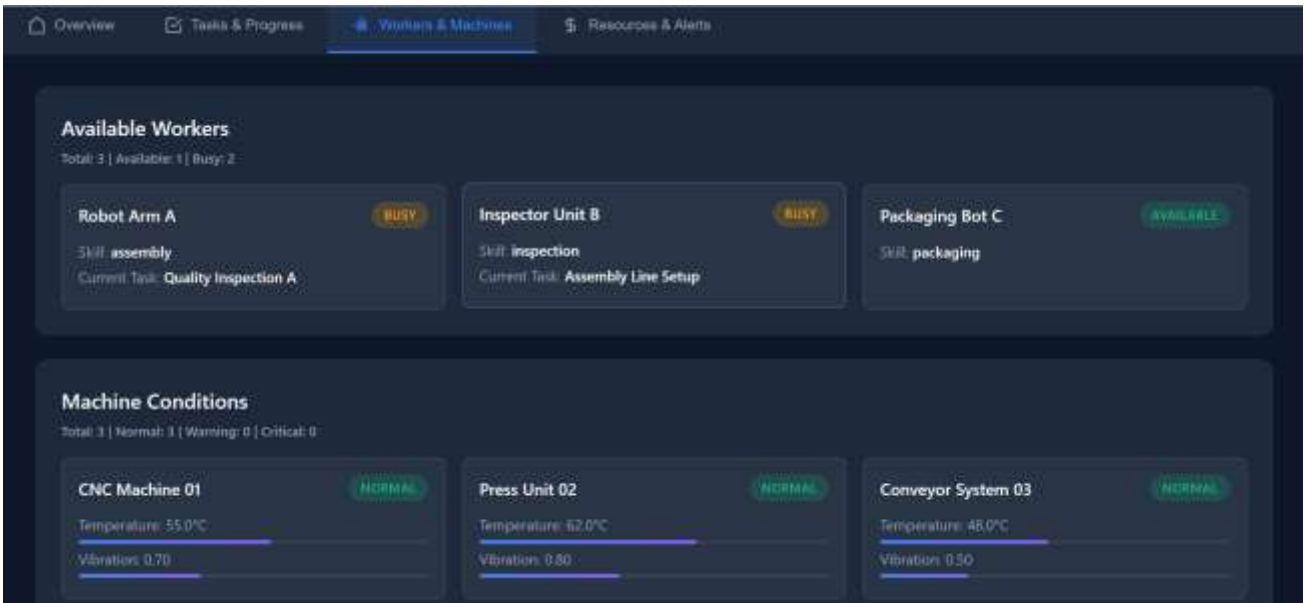
- **Simulation Controls:** Start, pause, next step, or reset the simulation.



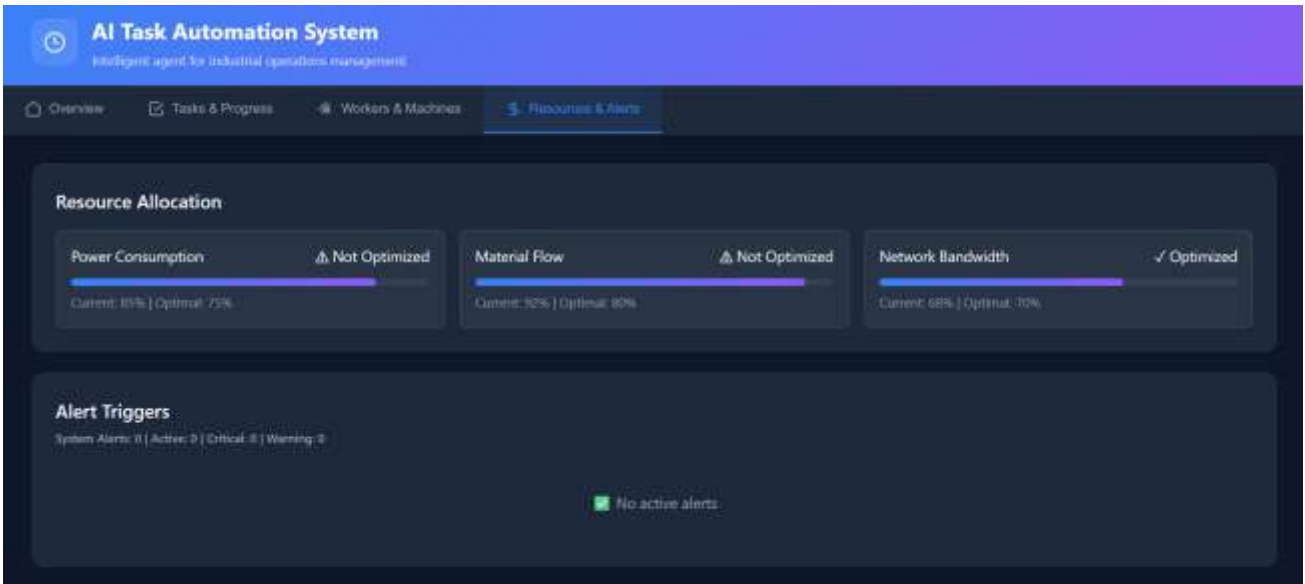
- **Task Management Panel:** Displays task status (Pending, In Progress, Completed) and progress percentage.



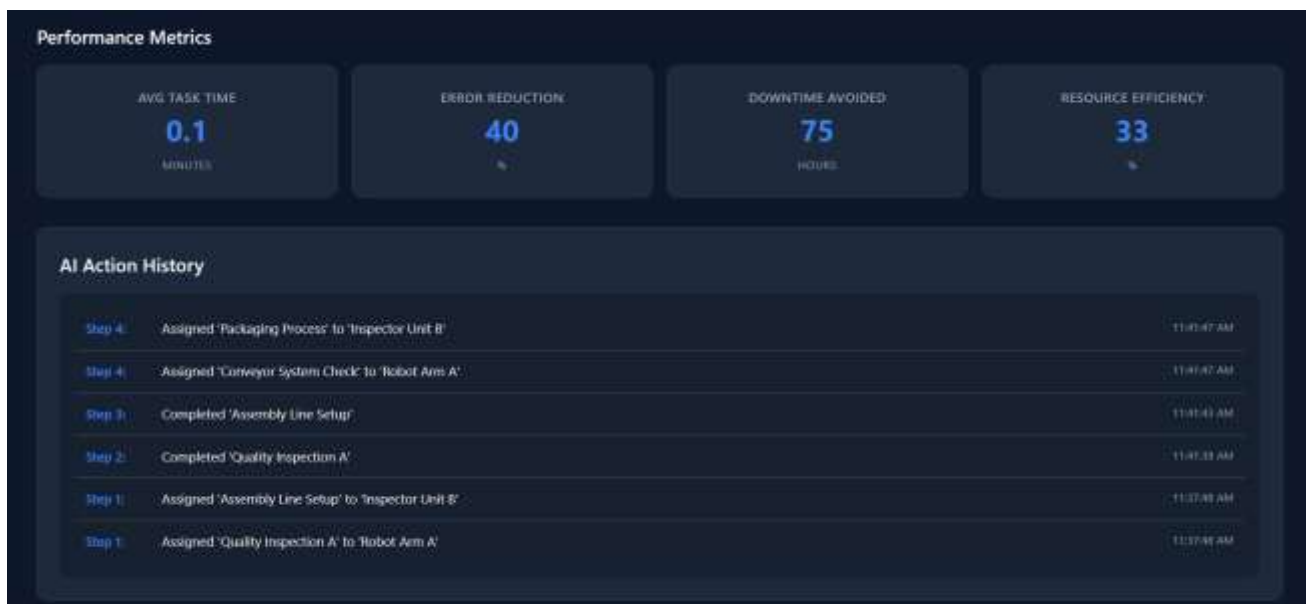
- **Resource View:** Shows worker availability and machine health states.



- **Alert Panel:** Displays critical machine alerts triggered by abnormal conditions.



- **Action History:** Logs all actions taken by the AI agent for transparency and evaluation.



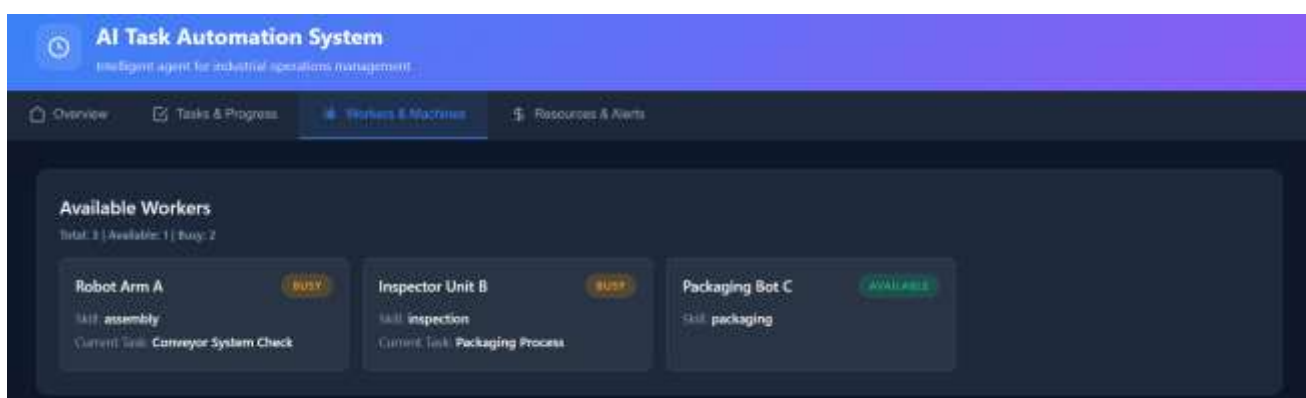
This UI ensures that the AI agent's reasoning and actions are observable, interpretable, and suitable for the project demonstration.

SYSTEM ALGORITHMIC LOGIC (A1-A3)

A1: Knowledge Representation – Assign Task to Worker/Machine

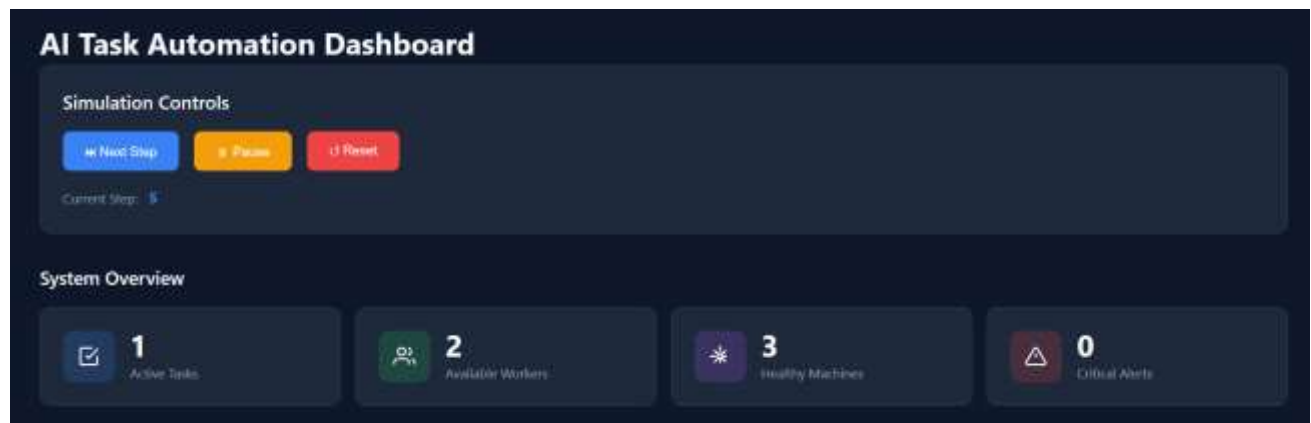
The AI agent continuously scans for pending tasks and matches them with available workers. When conditions are met, the agent assigns the task, updates task status to *In Progress*, and marks the worker as *Busy*.

This action optimizes workforce utilization and ensures continuous task execution.



A2: State Space Search – Trigger Dependent Tasks

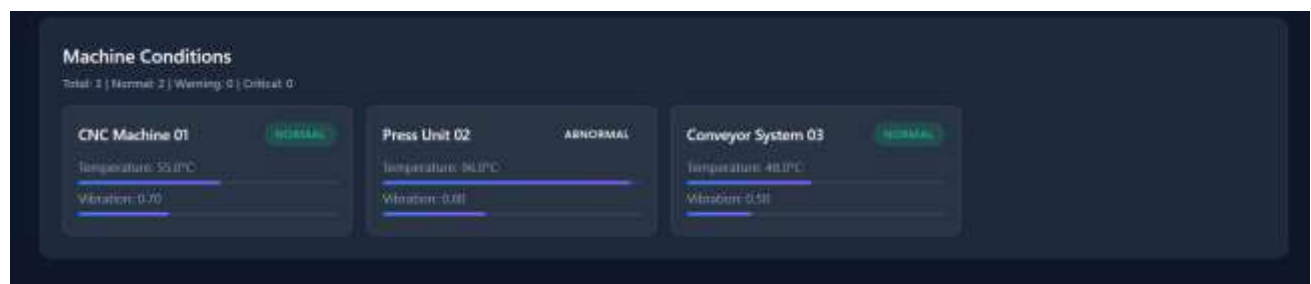
Once a task reaches 100% completion, the AI agent automatically triggers any dependent tasks. This guarantees workflow continuity and eliminates manual intervention, ensuring smooth progression across multi-step processes.



A3: PEASE – Monitor Machine and Schedule Maintenance

The agent monitors machine sensor data such as temperature. If thresholds are exceeded, the system changes the machine state to *Warning* or *Critical* and generates alerts. This simulates predictive maintenance, reducing the risk of system failure and operational downtime.

Together, these actions demonstrate a goal-oriented, rule-based AI agent that perceives, decides, and acts within its environment.



CONCLUSION

This Proof of Concept successfully demonstrates how an AI agent behaves intelligently using Actions A1–A3 within a task automation system. By integrating task allocation, dependency handling, and machine monitoring, the agent exhibits autonomous, rational, and adaptive behavior.

The PEAS-based design ensures that the agent’s performance is measurable, its environment is clearly defined, and its sensors and actuators are effectively simulated. Overall, the POC validates the feasibility of applying AI agents to real-world operational automation while remaining simple, explainable, and suitable for academic evaluation.

PEER REVIEW

Peer Review on Team Work	
Reviewer: Tahiya Nuzhath Khan	
Team Members	Category
Malika	4
Aria	4
Pashmia	4